

(d) Pairing of an initial antiparticle having quantum numbers $\mathbf{p}, \sigma, n^c$ with a field adjoint $\psi_\ell^\dagger(x)$ in $\mathcal{H}_i(x)$ yields a factor

$$\left[\psi_\ell^\dagger(x), a^\dagger(\mathbf{p} \sigma n^c) \right]_{\mp} = (2\pi)^{-3/2} e^{ip \cdot x} v_\ell^*(\mathbf{p} \sigma n). \quad (6.1.12)$$

(e) Pairing of a final particle (or antiparticle) having numbers $\mathbf{p}', \sigma', n'$ with an initial particle (or antiparticle) having quantum numbers $\mathbf{p}, \sigma, n$ yields a factor

$$\left[a(\mathbf{p}' \sigma' n'), a^\dagger(\mathbf{p} \sigma n) \right]_{\mp} = \delta^3(\mathbf{p}' - \mathbf{p}) \delta_{\sigma' \sigma} \delta_{n' n}. \quad (6.1.13)$$

(f) Pairing of a field $\psi_\ell(x)$ in $\mathcal{H}_i(x)$ with a field adjoint $\psi_m^\dagger(y)$ in $\mathcal{H}_j(y)$ yields a factor*

$$\begin{aligned} \theta(x - y) \left[\psi_\ell^+(x), \psi_m^{+\dagger}(y) \right]_{\mp} \pm \theta(y - x) \left[\psi_m^{-\dagger}(y), \psi_\ell^-(x) \right]_{\mp} \\ \equiv -i \Delta_{\ell m}(x, y), \end{aligned} \quad (6.1.14)$$

where ψ^+ and ψ^- are the terms in ψ that destroy particles and create antiparticles, respectively:

$$\psi_\ell^+(x) = (2\pi)^{-3/2} \int d^3p \sum_{\sigma} u_\ell(\mathbf{p} \sigma n) e^{ip \cdot x} a(\mathbf{p} \sigma n), \quad (6.1.15)$$

$$\psi_\ell^-(x) = (2\pi)^{-3/2} \int d^3p \sum_{\sigma} v_\ell(\mathbf{p} \sigma n) e^{-ip \cdot x} a^\dagger(\mathbf{p} \sigma n^c). \quad (6.1.16)$$

Recall that $\theta(x - y)$ is a step function, equal to +1 for $x^0 > y^0$ and zero for $x^0 < y^0$. These step functions appear in Eq. (6.1.14) because of the time-ordering in Eq. (6.1.1); we can encounter a pairing of an annihilation field $\psi^+(x)$ in $\mathcal{H}(x)$ with a creation field $\psi^{+\dagger}(y)$ in $\mathcal{H}(y)$ only if $\mathcal{H}(x)$ was initially to the left of $\mathcal{H}(y)$ in Eq. (6.1.1), i.e., if $x^0 > y^0$; similarly, we encounter a pairing of an annihilation field $\psi^{-\dagger}(y)$ in $\mathcal{H}(y)$ with a creation field $\psi^-(x)$ in $\mathcal{H}(x)$ only if $\mathcal{H}(y)$ was initially to the left of $\mathcal{H}(x)$ in Eq. (6.1.1), i.e., if $y^0 > x^0$. (The $\pm$ sign in the second term in (6.1.14) will be explained a little later.) The quantity (6.1.14) is known as a *propagator*; it is calculated in the following section.

The S -matrix is obtained by multiplying these factors together, along with additional numerical factors to be discussed below, then integrating over $x_1 \cdots x_N$, then summing over all pairings, and then over the numbers of interaction of each type. Before filling in all the details, it will be convenient first to describe a diagrammatic formalism for keeping track of all these pairings.

* If the interaction $\mathcal{H}(x)$ is written in the normal-ordered form, as in Eq. (5.1.33), then there is no pairing of fields and field adjoints in the *same* interaction. Otherwise some sort of regularization is needed to give meaning to $\Delta_{\ell m}(0)$.